# Mineral Resources

**Definition of Terms**

- **Resource** - stock or supply of anything that can be drawn on by anyone to function effectively
- **Mineral** - a solid, inorganic substance of natural occurrence

**Characteristics**

1. **Naturally Occurring** - minerals are made, and exist, naturally
2. **Inorganic** - minerals aren't made from living organisms
3. **Solid** - minerals exist in neither liquid nor gas forms
4. **Definite chemical composition** - minerals are defined by their chemical composition, which must be expressed by their chemical formula in specific ratios
5. **Ordered internal structure** - minerals must have regular, repetitive geometric patterns or crystal structures

**Mineral Classifications**

I. **Composition**

1. **Silicates** composed primarily of silicon-oxygen tetrahedrons ($SiO_4{}^{2-}$)
   **e.g.**
   Olivine ($(Mg, Fe)_2SiO_4$)



*Source:* https://en.wikipedia.org/wiki/Olivine

Quartz ($SiO_2$)



*Source:* https://www.minerals.net/mineral/quartz.aspx

2. **Oxides** consist of metal cations bonded to oxygen anions ($O^{2-}$)
   **e.g.**
   Magnetite ($Fe_3O_4$)



*Source:* https://en.wikipedia.org/wiki/Magnetite

Hematite ($Fe_2O_3$)



*Source:* https://en.wikipedia.org/wiki/Hematite

3. **Sulfides** consist of metal cations bonded to sulfides ($S^{2-}$); commonly found mineral alongside oxides since metals are found in large proportions in the minerals
   **e.g.**
   Galena ($PbS$)



*Source:* https://www.britannica.com/science/galena-mineral

Pyrite ($FeS_2$)



*Source:* https://www.britannica.com/science/pyrite

4. **Sulfates** consist of metal cations bonded to sulfate ($SO_4{}^{2-}$) anionic groups; usually precipitate out of the water near the Earth's surface
   **e.g.**
   Gypsum ($CaSO_4 \cdot 2H_2O$)



*Source:* https://en.wikipedia.org/wiki/Gypsum

Barite / Baryte ($BaSO_4$)



*Source:* https://en.wikipedia.org/wiki/Baryte

5. **Halides** consist of metal cations bonded to halogens
   **e.g.**
   Halite / Rock salt ($NaCl$)



*Source:* https://en.wikipedia.org/wiki/Halite

Fluorite ($CaF_2$)



*Source:* https://geology.com/minerals/fluorite.shtml

6. **Carbonates** consist of minerals with carbonic ion present that easily bond with metals
   **e.g.**
   Calcite ($CaCO_3$)



*Source:* https://geology.com/minerals/calcite.shtml

Dolomite ($CaMg(CO_3)_2$)



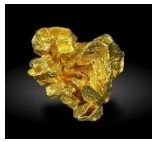
*Source:* https://en.wikipedia.org/wiki/Dolomite_(mineral)

7. **Natives** / **Native metals** consist of singular metal elements
   **e.g.**
   Copper ($Cu$)



*Source:* https://geology.com/minerals/copper.shtml

Gold ($Au$)



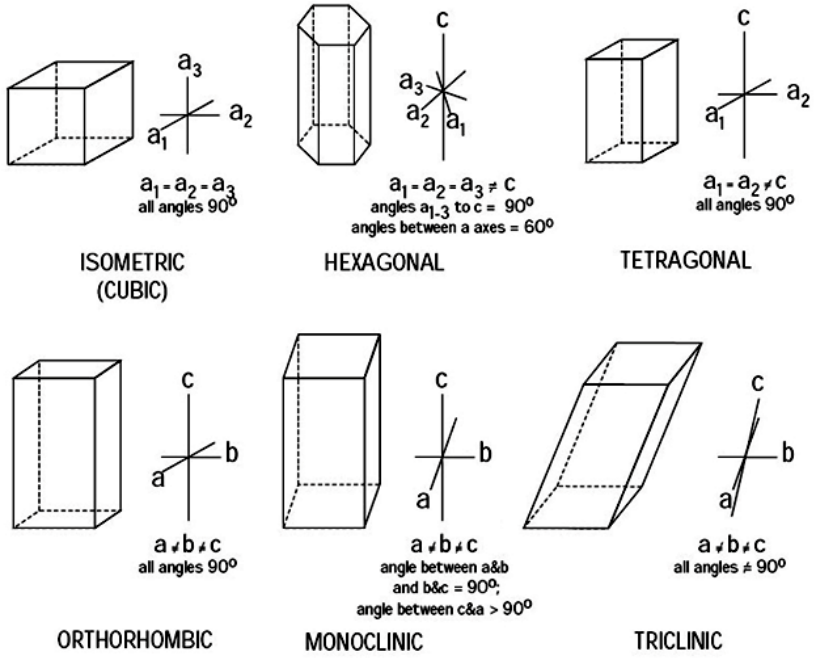
*Source:*
https://www.fabreminerals.com/search_show.php?SECTION=RSJC&CODE=TM88AG8

## II. Crystal Structure

- formation of the mineral according to its chemical composition




*Source:* https://www.luckysci.com/2014/01/best-trick-for-learning-the-seven-crystal-systems/

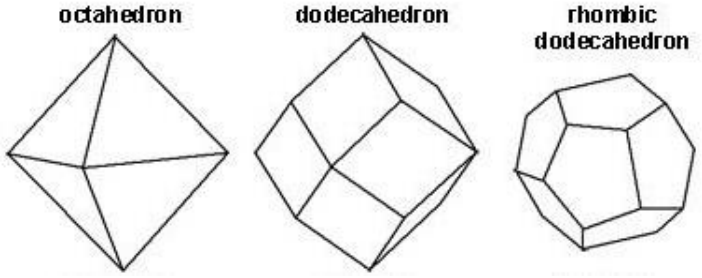



*Source:* https://www.luckysci.com/2014/01/best-trick-for-learning-the-seven-crystal-systems/

**DETAILED Structure**

1. **Isometric**
   - common crystal system whose unit cell is in the shape of a cube $(a_1, a_2, a_3)$

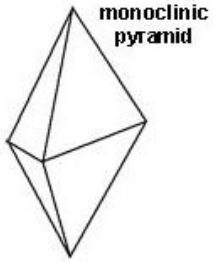
   **Other forms:**




*Source:* http://geologycafe.com/gems/chapter2.html

2. **Monoclinic**
   - crystal structure whose axes [(x, y, z) ***OR*** (a, b, c)] have different lengths; both x- and z-axes meet at 90° while the y-axis does not

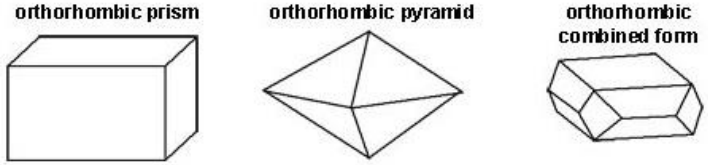
   **Other form:**




*Source:* http://geologycafe.com/gems/chapter2.html

3. **Orthorhombic**
   - crystal structure whose axes [(x, y, z) ***OR*** (a, b, c)] have different lengths; all three (3) axes meet at 90°

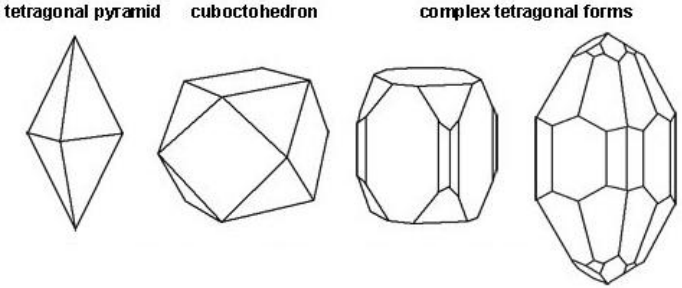
   **Other forms:**




*Source:* http://geologycafe.com/gems/chapter2.html

4. **Tetragonal**
   - crystal structure whose two (2) axes [($x_1$, $x_2$) ***OR*** ($a_1$, $a_2$)] have equal lengths while the third axis ($z$ ***OR*** $c$) is longer; all three (3) axes meet at 90°

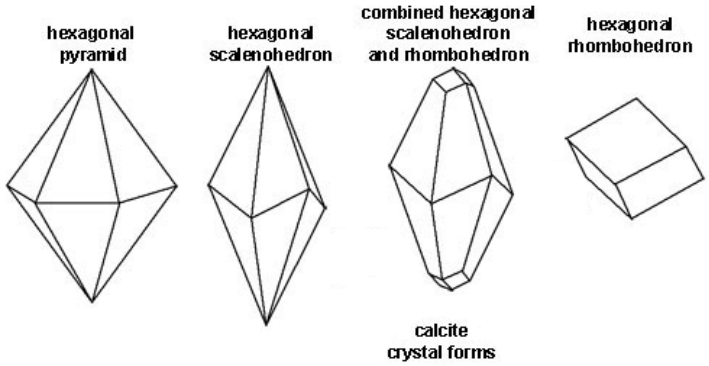
   **Other forms:**




*Source:* http://geologycafe.com/gems/chapter2.html

5. **Hexagonal**
   - has four (4) crystallographic axes consisting of three (3) equilateral axes [($x_1$, $x_2$, $x_3$) ***OR*** ($a_1$, $a_2$, $a_3$)] angling at 120° to each other, with one (1) perpendicular axis ($z$ ***OR*** $c$)
   - trigonal formations are categorized here

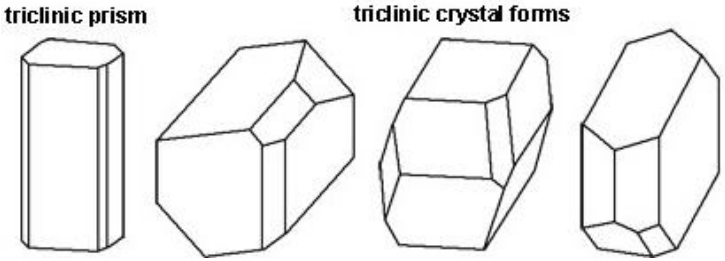
   **Other forms:**




*Source:* http://geologycafe.com/gems/chapter2.html

6. **Triclinic**
   - aka. The rhombohedron system
   - crystal system that is similar to the isometric system, but it has been skewed to one (1) side, making it oblique

   **Other forms:**

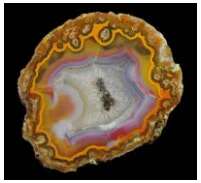


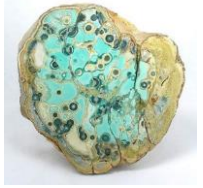

*Source:* http://geologycafe.com/gems/chapter2.html

## III. Habit

- the typical outward appearance of a mineral's crystal form

| HABIT Type | Description | Example |
|---|---|---|
| Acicular | • Crystals have a needle-like shape that ends in a blunt termination.<br>• Can cluster to form fan-shaped or radially-shaped aggregates<br>• Used when an individual crystal's length is greater than its width or diameter |  |
| Banded | • Have narrow layers (bands) of different color and/or texture<br>• May be a response to changes in the composition of the mineral's growth, the sedimentary process, or other conditions |  |
| Bladed | • Elongated crystals whose length exceeds the width and its width exceeds the mineral's thickness<br>• Ends sometimes taper to a point<br>• Shaped similarly to a sword or knife blade |  |
| Botryoidal | • aka. *globular* or *mammillary*<br>• Gk. *botruoeides*, "a bunch of grapes"<br>• Mineral aggregates are round |  |
| Columnar | • Long prisms whose widths exceeds the acicular type<br>• May contain multiple parallel crystals in one "column" |  |
| Cubic | • Crystals with six square faces and four-fold rotational symmetry around three (3) axes |  |
| Dendritic | • Minerals that form a branching pattern similar to a tree or a leaf vein<br>• Gk. *dendron*, "tree" |  |
| Dodecahedral | • Minerals that have 12 flat faces |  |
| Drusy | • Surface is covered in small crystals (druse) |  |
| Fibrous | • Minerals occur in very fine, fiber-like crystals, reminiscent of hair or fur<br>• Thinner than acicular crystals<br>• Aggregates similarly with acicular crystals |  |

| | | |
|---|---|---|
| Foliated | • aka. *micaceous*, as most mica minerals have this habit<br>• Sheet-like crystal habit<br>• Minerals with this habit can be cut into thin sheets |  |
| Geodic | • Minerals aggregate in rounded or oblate masses through crystallization on the inside walls of a hollow cavity<br>• Bands eventually develop, gradually filling in the cavity without filling it completely<br>• These aggregates are called *geodes* |  |
| Granular | • Aggregations made of many rounded or equant anhedral crystals of similar sizes<br>• Can be loose without any interstitial material, or could be interlocking with other materials |  |
| Hopper | • Partially-formed crystals that have experienced more rapid growth on the outer edges than in the center<br>• Well-developed outer sides, "hollow" center<br>• Can be considered as an "innie" |  |

| | | |
|---|---|---|
| Massive | • Crystals with no distinctive geometry or habit |  |
| Nodular | • Crystals grow to form rounded/ bulbous structures<br>• Usually arranged in radial structures within the nodule<br>• Nodule may experience concentric banding |  |
| Octahedral | • Crystals form into an octahedral shape<br>• Octahedrons have eight (8) triangular faces and three (3) axes of four-fold rotational symmetry |  |
| Oolitic | • Occur in crystalline aggregates that are rounded and less than 4 mm in size<br>• Smaller version of pisolites |  |
| Pisolitic | • Occur in crystalline aggregates that are rounded and about the size of peas<br>• Individual pisolites are made of many tiny radiating mineral crystals |  |
| Prismatic | • Minerals that form in elongated crystals with opposite faces usually parallel to each other |  |

| | | |
|---|---|---|
| | • Often striated either along the length or across the width | |
| Radiating | • Aggregates grow outwards from a central point<br>• Consist of multiple crystals growing in diverging directions |  |
| Rosette | • Clusters of tabular crystals in a radial arrangement that have external geometry resembling a rose or a flower |  |
| Stalactitic | • Used for specimens that formed as stalactites or stalagmites<br>• Often grow downwards or upwards in a cavity or a cavern<br>• Still have radial cross-sections |  |
| Striated | • *Striations* are fine, slightly indented lines present on the faces of some crystals<br>• Striations are always parallel to the crystallographic axis and one (1) edge of the crystal face |  |
| Tabular | • Flat and plate-like<br>• Length and width are larger than the thickness<br>• Known as stone tablets |  |

*Sources:* https://geology.com/minerals/crystal-habit/

**Mineral Splitting**

- **Cleavage** - the tendency of minerals to break along the planes of weakness
  - splitting between two (2) atomic planes
  - result of weaker bond strengths or greater lattice spacing across the plane in question

- **Fracture** - the resulting surface of a mineral after experiencing cleavage and/or parting
  - can also result from breaking a mineral outside its cleavage points

- **Parting** - results from applying stress to a mineral
  - the tendency of minerals to break along planes of structural weakness (similar to cleavage), but is due to external stress
  - also results from breaking along the planes of twinned minerals (i.e., minerals that have been "fused" together)

**KINDS OF MINERAL CLEAVAGE**

| HABIT Type | Description |
|---|---|
| Eminent | Mineral cleaves readily and very easily |
| Perfect | Mineral surface is smooth and mirror-like |
| Distinct | Mineral surface is smooth with minor inconsistencies (e.g., visible cracks, rough edges and/or exterior) |
| Difficult | Mineral surface contains inconsistencies after cleavage, may tend to fracture than to cleave |
| Indistinct | Mineral surface is present but is barely noticeable |
| None | No cleavage available |

**CLEAVAGE HABITS**

| HABIT Type | Description and Cleavage Direction |
|---|---|
| Basal / Pinacoidal | Cleaves at the base, one (1) direction |
| Cubic | Cleaves as cubes, three (3) directions at right angles |
| Dodecahedral | Cleaves as dodecahedrons, six (6) directions |
| Octahedral | Cleaves as octahedrons, four (4) directions |
| Prismatic | Cleaves to form prisms, two (2) directions |
| Rhombohedral | Cleaves into rhombohedrons, three (3) directions not at right angles |

## FRACTURE HABITS

| HABIT Type | Description |
|---|---|
| Conchoidal | • Sometimes referred to as a *shelly*<br>• Fracture resembles a semicircular shell with a smooth, curved surface<br>• Can be seen in broken glass |
| Uneven | • Fracture leaves a rough or irregular surface |
| Hackly | • aka. *Jagged* fracture<br>• Fracture resembles a broken metal<br>• Fracture has rough, jagged points<br>• Seen in true metals |
| Splintery | • Fracture forms elongated splinters<br>• Exhibited by fibrous minerals |
| Earthy | • aka. *Crumbly* fracture<br>• Mineral crumbles upon fracture, reminiscent of dry earth |
| Smooth | • aka. *Even* fractures<br>• Fracture with a smooth surface reminiscent of a perfect cleavage |
| Subconchoidal | • Cross between a conchoidal and smooth fractures<br>• Fracture with a smooth, yet irregular rounded corners |

## Luster/Lustre

- property of a mineral to reflect light
- describes how brilliant or dull it is

| Luster Type | Description | Example |
|---|---|---|
| Adamantine | • Transparent or translucent minerals with very high refractive indices<br>• Gives off brilliance or shine to the minerals | Cerussite, Diamond |
| Earthy / Dull | • Exhibits very poor luster<br>• Observed from minerals with rough and/or porous surfaces | Goethite, Kaolinite |
| Greasy | • Has a dull sheen that resembles fat, oil, or it may look like it has been dipped and smoothened in oil<br>• May contain microscopic inclusions<br>• May have a "greasy" feel | Graphite, Nephrite |
| Metallic | • Opaque and reflective sheen<br>• Resembles polished metals | Magnetite, Pyrite |
| Pearly | • Resembles a pearl in sheen<br>• Consists of thin coplanar sheets that breaks up light in various layers | Labradorite, Muscovite |
| Resinous | • Resembles resin, chewing gum, or smooth plastic<br>• Has a "honey-like" sheen, with a "plastic" feel | Sulfur, Wulfenite |
| Silky | • Minerals contain fibrous structures<br>• Resembles silk cloth and/or fiber in terms of optical properties | Asbestos, Serpentine |
| Submetallic | • Minerals are opaque to semi-opaque<br>• Sheen almost looks similar to metal<br>• Thin splinters of minerals with this luster are translucent | Hornblende, Rutile |
| Vitreous | • Almost 65% of minerals have this luster<br>• Minerals have glass-like reflective properties | Quartz, Sugilite |
| Waxy | • Minerals appear to be "coated in wax"<br>• Has a "waxy" feel | Chrysoprase, Turquoise |

**Specific Gravity**

Specific gravity is the measure of a mineral's density. It is the weight of a mineral relative to the weight of an equal volume of water. Since water has a specific gravity of one (1), any material whose specific gravity is greater than one (1) is denser than water. Likewise, materials with specific gravities lower than one (1) are less thick.

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